

LSTM Model :

On the validation set the LSTM model shows good and consistent performance on the classes least affected by imbalance and with larger support toxic achieved F1 = 0.78, obscene F1 = 0.80, and insult F1 = 0.70.

In contrast, performance drops noticeably on the classes suffering from stronger imbalance and fewer samples: threat (only 110 samples) achieved F1 = 0.42, identity_hate (281 samples) F1 = 0.45, and severe_toxic (320 samples) F1 = 0.52.

F1 Micro = 0.74

F1 Weighted Avg = 0.74

	precision	recall	f1-score	support
toxic	0.79	0.78	0.78	3059
severe_toxic	0.45	0.63	0.52	320
obscene	0.79	0.82	0.80	1690
threat	0.44	0.41	0.42	110
insult	0.65	0.76	0.70	1575
identity_hate	0.42	0.49	0.45	281
micro avg	0.71	0.76	0.74	7035
macro avg	0.59	0.65	0.61	7035
weighted avg	0.72	0.76	0.74	7035
samples avg	0.06	0.07	0.06	7035

On the test set performance was poor clearly across all classes compared to validation. toxic (support 6090) achieved F1 = 0.60, obscene (support 3691) F1 = 0.61, and insult (support 3427) F1 = 0.55, noticeably lower than their validation scores.

The imbalanced classes suffer even more: threat (211 samples) F1 = 0.30, severe_toxic (367 samples) F1 = 0.37, and identity_hate (712 samples) F1 = 0.46.

This overall decline is largely due to the different nature and quality of the test data it originates from the Kaggle Toxic Comment Classification competition's official test set which is known to be considerably harder and noisier than the training distribution .

Overall test scores:

F1 Micro = 0.57

F1 Weighted Avg = 0.57

	precision	recall	f1-score	support
toxic	0.46	0.85	0.60	6090
severe_toxic	0.25	0.71	0.37	367
obscene	0.49	0.80	0.61	3691
threat	0.26	0.35	0.30	211
insult	0.43	0.77	0.55	3427
identity_hate	0.41	0.53	0.46	712
micro avg	0.45	0.79	0.57	14498
macro avg	0.38	0.67	0.48	14498
weighted avg	0.45	0.79	0.57	14498
samples avg	0.07	0.07	0.07	14498

RNN Model :

On the validation set the RNN (SimpleRNN) model shows the same pattern seen with the LSTM: strong performance on the classes with less imbalance and larger support toxic F1 = 0.78, obscene F1 = 0.80, and insult F1 = 0.70. In

contrast, performance drops noticeably on the classes with stronger imbalance: threat (110 samples) achieved F1 = 0.33 only, severe_toxic (320 samples) F1 = 0.50, and identity_hate (281 samples) F1 = 0.47.

Overall validation scores:

F1 Micro = 0.73

F1 Weighted Avg = 0.74

	precision	recall	f1-score	support
toxic	0.82	0.76	0.78	3059
severe_toxic	0.39	0.69	0.50	320
obscene	0.81	0.79	0.80	1690
threat	0.27	0.41	0.33	110
insult	0.63	0.79	0.70	1575
identity_hate	0.45	0.49	0.47	281
micro avg	0.71	0.75	0.73	7035
macro avg	0.56	0.65	0.60	7035
weighted avg	0.73	0.75	0.74	7035
samples avg	0.06	0.07	0.06	7035

And the performance on the test set was poor.

F1 Micro = 0.61

F1 Weighted Avg = 0.62

	precision	recall	f1-score	support
toxic	0.52	0.85	0.64	6090
severe_toxic	0.21	0.72	0.32	367
obscene	0.62	0.76	0.68	3691
threat	0.18	0.51	0.27	211
insult	0.45	0.80	0.57	3427
identity_hate	0.48	0.60	0.53	712
micro avg	0.49	0.79	0.61	14498
macro avg	0.41	0.71	0.50	14498
weighted avg	0.51	0.79	0.62	14498
samples avg	0.07	0.07	0.07	14498